

CHAPTER-1

INTRODUCTION

1. INTRODUCTION

1.1 Background of Infant Safety & Monitoring Systems

Infant safety is one of the most critical concerns for parents and caregivers, especially during the early stages of life. Babies are highly vulnerable to risks such as suffocation due to unsafe sleeping positions, prolonged crying, wet cradle conditions, and unfavourable environmental changes (temperature or humidity). Traditionally, infants are monitored manually by parents or nurses, which requires continuous attention and causes stress for caregivers.

With the advancement of Internet of Things (IoT) and Artificial Intelligence (AI), smart cradle monitoring systems have emerged as a promising solution. IoT enables real-time data collection from sensors and remote monitoring through mobile devices, while AI provides predictive alerts by analysing patterns of posture, crying, and environmental changes. Such intelligent monitoring reduces the dependency on human supervision and ensures timely intervention to prevent health risks.

1.2 Problem Statement

Despite the availability of modern baby care equipment, most existing cradles lack real-time intelligent monitoring and predictive alert systems. Conventional monitoring devices often:

- Fail to detect unsafe sleep positions that may lead to suffocation.
- Do not provide predictive alerts before a problem escalates.
- Require constant manual observation, increasing caregiver stress.
- Have limited remote monitoring capability, especially in homes and neonatal care units.

1.3 Motivation for Choosing this Project

The motivation behind this project is the growing concern of infant mortality and safety risks due to lack of timely detection of unsafe conditions. According to health statistics, a significant percentage of infant deaths are caused by Sudden Infant Death Syndrome (SIDS) and unattended accidents.

As engineering students, we are motivated to apply IoT and AI technologies to create a smart cradle system that not only monitors infants but also provides predictive, proactive, and remote safety solutions. This project also bridges the gap between theoretical learning and real-world application, preparing us for solving socially relevant problems using engineering and innovation.

1.4 Objectives of the Project

- To develop a real-time cradle monitoring system using IoT and AI.
- To detect unsafe infant sleep positions using vision-based analysis.
- To monitor environmental and behavioural conditions using multiple sensors (temperature, humidity, wetness, sound).
- To provide predictive safety alerts to caregivers via Telegram/mobile notifications.
- To automate cradle responses such as fan activation (for comfort) and servo-based movement (to calm the baby).
- To design a scalable, low-cost, and reliable solution suitable for both home and hospital use.

1.5 Scope and Limitations

1.5.1 Scope

- Provides **real-time infant safety monitoring** using IoT and AI.
- Sends **instant alerts to caregivers** through cloud and mobile applications.
- Automates cradle operations (swinging, fan control) for baby comfort.
- Can be extended to **neonatal intensive care units (NICUs)** for hospital-level safety.

1.5.2 Limitations

- Requires **continuous power supply**; system may fail during power/battery outage.
- Internet connectivity is essential for **cloud-based monitoring and AI analysis**.

- Limited to the **accuracy of sensors and AI models** (e.g., false positives/negatives in posture or crying detection).
- Current version does not include **advanced medical monitoring** (heartbeat, SpO₂), which can be considered in the future scope.

CHAPTER-2

LITERATURE SURVEY

2. LITERATURE SURVEY

AI-enhanced cradle monitoring systems have gained importance in recent years due to the increasing need for continuous infant safety, health monitoring, and timely caregiver intervention. Technologies such as the Internet of Things (IoT), embedded sensors, computer vision, and artificial intelligence (AI) enable real-time monitoring of infant movement, crying, and environmental conditions. This chapter reviews existing research on IoT-based baby monitoring, AI-driven cry and posture analysis, and automated alert systems to identify their strengths and limitations. The review helps justify the need for an AI-enhanced cradle monitoring system with predictive infant safety alerts to improve accuracy, responsiveness, and overall infant well-being.

2.1 Survey of Existing System

This work presents an IoT-based smart cradle that monitors baby movement, crying sound, and environmental comfort using sensors integrated with a microcontroller. Real-time data is transmitted to caregivers for timely response. The study shows improved monitoring accuracy and quick alerts, but the system lacks advanced AI prediction and depends on stable internet connectivity [1].

The authors developed a smart cradle integrated with IoT and a mobile application to enable remote monitoring and parental control. The system enhances monitoring efficiency and control latency, allowing caregivers to manage cradle functions remotely. However, the solution is constrained by power dependency and network reliability issues [2].

This research combines IoT sensors with machine learning techniques to recognize infant emotions using cry sounds and facial expressions. The system demonstrates high detection accuracy and improved understanding of infant needs. Its major limitation is the requirement of large datasets and higher computational resources [3].

The study focuses on detecting infant crying using audio feature extraction and machine learning classifiers in real-life conditions. Results indicate strong sensitivity and specificity even in noisy environments. However, background noise and varying acoustic conditions still affect performance [4].

This paper introduces an innovative neonatal monitoring system using IoT sensors embedded in smart textiles to track physiological parameters. The system provides accurate

real-time alerts for neonatal safety. Despite its effectiveness, high implementation cost and limited scalability remain challenges [5].

The authors applied transfer learning on infant cry signals to diagnose perinatal asphyxia. The study proves that deep learning can extract meaningful health indicators from cries with high accuracy. However, the approach requires labeled datasets and expert model training [6].

This research focused on classifying infant pain levels using machine learning-based cry analysis. The method involved feature extraction and supervised learning algorithms to assess discomfort. Results showed effective pain classification, reinforcing the role of AI in infant monitoring systems. However, the approach required expert-labeled data and lacked integration with IoT platforms for real-time alerts [7].

The authors presented an open-source framework for extracting meaningful audio features from infant cry signals using signal processing and machine learning techniques. The key contribution was providing standardized tools that enable predictive analysis of infant conditions. Findings supported reliable cry-based pattern recognition. The main limitation was that the system depended heavily on audio quality and did not integrate environmental or visual data [8].

This study applied deep learning techniques to detect and classify baby cries in real domestic environments using audio signal processing. The method involved extracting audio features and training neural networks for cry recognition. Results showed high accuracy in identifying distress cries, supporting automated alert systems. However, the approach required large labeled datasets and was sensitive to background noise [9].

The paper introduced a cloud-centric IoT healthcare architecture where sensor data is processed and stored on cloud platforms. The method used IoT nodes with cloud services for monitoring and alert generation. Key findings demonstrated improved accessibility and real-time health monitoring through cloud dashboards. Limitations include high dependency on internet connectivity and the absence of AI-based behaviour prediction [10].

This work proposed an early intelligent baby monitoring system using basic sensors such as temperature and sound sensors combined with wireless communication. The method focused on continuous sensing and remote data transmission to caregivers. Key

findings showed that real-time monitoring improves infant safety and reduces manual supervision. However, the system lacked cloud integration, advanced analytics, and AI-based prediction, limiting its scalability and intelligence [11].

An overview of the related research works, highlighting their methodologies and key outcomes, is presented in Table 2.1

Table 2.1 Survey Existing System

S.No	Title	Method Used	Key Findings	Limitations
1	IoT-Based Baby Cradle Monitoring System	IoT sensors with ESP-based controller	Improved baby monitoring accuracy and faster alert response	Dependent on sensor quality and internet connectivity
2	IoT-Enabled Smart Baby Cradle for Enhanced Infant Monitoring	IoT cradle with mobile app integration	Enabled remote parental control and efficient monitoring	Network latency and power dependency
3	IoT-Based Baby Monitoring System with Emotion Recognition	IoT sensors + ML-based emotion recognition	Accurate emotion detection from cry and facial data	Requires large datasets and higher computation
4	Infant Crying Detection in Real-World Environments	Audio feature extraction + ML classification	Reliable cry detection in noisy environments	Performance affected by background noise
5	Baby-Guard: IoT-based Neonatal Monitoring with Smart Textiles	Smart textile sensors + IoT platform	Continuous physiological monitoring with real-time alerts	High cost and limited scalability
6	Neural Transfer Learning for Cry-based Diagnosis (2019)	Deep learning + transfer learning on cry signals	Effective health diagnosis from infant cries	Needs labeled datasets and expert training

7	Automatic Cry Analysis and Classification for Infant Pain Assessment	Machine learning-based cry classification	Successfully classified infant pain levels from cry patterns	Limited generalization across different infants and environments
8	Open-Source Feature Extraction Tool for Infant Cry Analysis	Signal processing + ML-based cry feature extraction	Provided standardized features for predictive infant cry analysis	Requires large labeled datasets and expert tuning
9	Baby Cry Detection in Domestic Environment Using Deep Learning	Audio feature extraction + deep neural networks	Achieved accurate baby cry detection in real-world environments	High computational cost, sensitive to background noise
10	Cloud-Centric IoT Based Healthcare Monitoring Framework	IoT sensors + cloud architecture + remote monitoring	Demonstrated scalable cloud-based health monitoring suitable for real-time alerts	Not baby-specific, lacks vision and emotion analysis
11	Design of an Intelligent Baby Monitoring System Using Sensors and Wireless Technology	Temperature sensor + sound sensor + wireless data transmission	Enabled continuous remote infant monitoring and laid the foundation for sensor-based baby monitoring systems	No cloud support, no AI analysis, limited alert intelligence

CHAPTER-3

EXISTING METHODOLOGY

3. EXISTING METHODOLOGY

3.1 Introduction

Existing IoT-based baby cradle monitoring systems focus mainly on basic monitoring using low-cost sensors and microcontrollers such as Arduino UNO. These systems usually track only simple parameters like crying sound, temperature, and Wetness, and send alerts through platforms like Blynk Cloud. Although these systems provide remote monitoring, they lack intelligence, automation, and predictive safety features. Most existing solutions do not use camera-based monitoring or machine learning, making them limited to reactive responses rather than early prediction of infant distress.

3.2 Block Diagram of Existing System

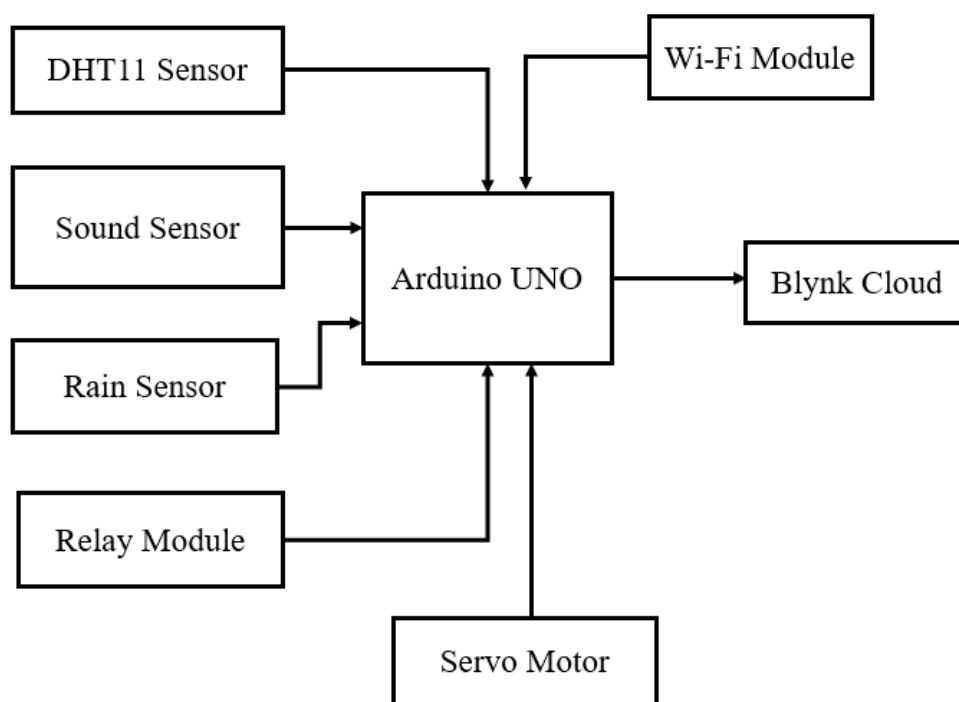


Figure 3.2 Flowchart of Existing System

Fig 3.2 represents the block diagram of an IoT-based cradle monitoring system using an Arduino UNO as the central controller. Various sensors including the DHT11 sensor for temperature and humidity, the sound sensor for detecting the baby's cry, and the rain sensor for monitoring moisture are connected to the Arduino to continuously collect environmental and behavioural data. The relay module is also interfaced with the Arduino

to control external devices such as a fan or other actuators. A servo motor is connected to the Arduino to enable automatic cradle movement whenever necessary, such as when the baby cries. All the collected data is sent from the Arduino UNO to the Blynk Cloud, allowing caregivers to monitor real-time conditions remotely through a mobile application. This system helps ensure the baby's comfort and safety by enabling automated cradle actions along with remote monitoring.

3.3 Flowchart of Existing System

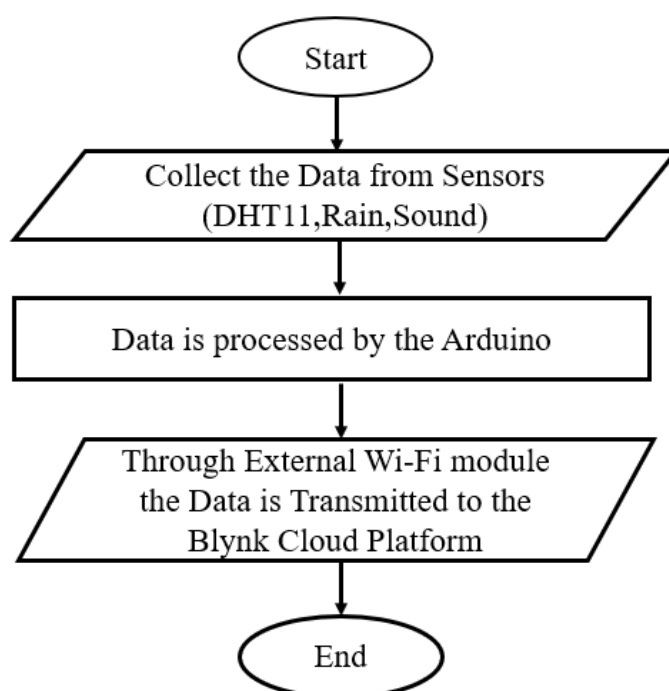


Figure 3.3 Flowchart of Existing System

The Fig 3.3 illustrates the working process of an IoT-based monitoring system using Arduino UNO. The operation begins with the system collecting environmental and condition-related data from the connected sensors, including the DHT11 sensor for temperature and humidity, the rain sensor, and the sound sensor. Once the data is gathered, it is transmitted to the Blynk Cloud through the Arduino UNO, enabling real-time monitoring on a mobile or web dashboard. After the data reaches the cloud, the system analyses the sensor readings and, whenever an unusual or critical condition is detected. This ensures immediate communication and quick response. Finally, the process ends after the successful delivery of alerts and monitoring data.

3.4 Working Principle

1. **Sensor Data Collection:** The system continuously collects data from sensors such as the sound sensor (cry detection), DHT11 temperature-humidity sensor, and motion sensors.
2. **Microcontroller Processing:** Arduino UNO processes the sensor values. If crying is detected, or if temperature/humidity crosses a threshold, specific actions are triggered.
3. **Cradle Automation:** If crying is detected, the motor is activated to gently swing the cradle automatically.
4. **Data Transmission:** The processed data is transmitted to Blynk Cloud, where parents can view the baby's condition remotely.
5. **Continuous Monitoring:** The system runs continuously to check environmental comfort and baby activity.

3.5 Existing Output

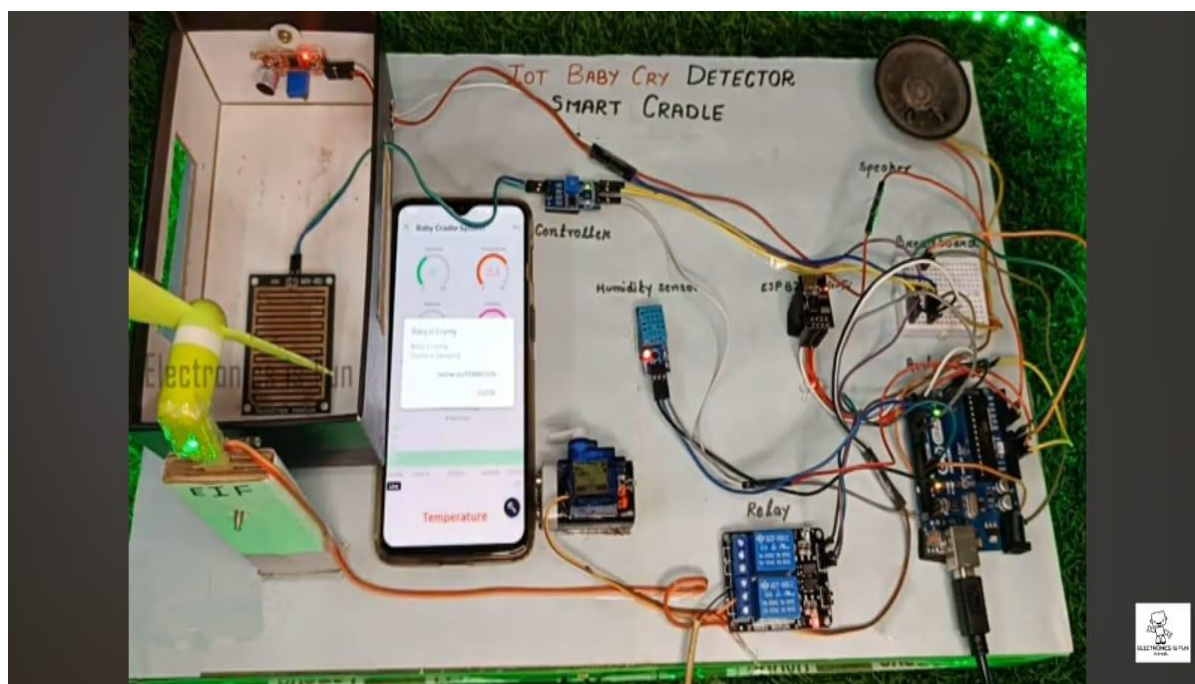


Figure 3.5 Existing Output

Fig. 3.5 illustrates the existing output of a basic baby-cry detector-based smart cradle system. The setup includes a combination of sensors such as the sound sensor, humidity and temperature sensor (DHT11), and a rain sensor detection module, all connected to an Arduino controller. The system interfaces with a mobile application to

display real-time readings and send alerts when the baby cries. A relay module controls external components like a fan or motor, while a small servo mechanism helps in demonstrating cradle movement. This prototype represents the current functionality of the existing system, showing its ability to detect baby cries, monitor environmental conditions, and trigger simple automated responses. The system can successfully monitor crying, temperature, humidity, and movement.

3.6 Limitations of the Existing System

- **No AI or predictive capability:** It cannot predict safety risks (e.g., suffocation, posture issues).
- **No camera monitoring:** The system cannot visually detect if the baby's face is covered or baby is in an unsafe position.
- **Limited sensors:** Only basic sensors are used; important parameters like wetness or detailed motion are missing.
- **Connectivity issues:** The system stops sending data if WiFi disconnects, making it unreliable.
- **No advanced alerting system:** Only Blynk app alerts are provided; no Telegram, no SMS-based emergency alerts.
- **Low automation:** System reacts only when the baby cries; no continuous smart regulation.
- **Arduino limitations:** Arduino UNO has limited processing power and no support for AI or camera modules.

CHAPTER-4

PROPOSED METHODOLOGY

4.PROPOSED METHODOLOGY

4.1 Introduction to the Proposed Idea

The proposed system enhances traditional cradle monitoring by integrating AI-based infant safety prediction with IoT-enabled real-time monitoring. Unlike existing cradle systems that rely on basic sensing (such as cry, motion, or temperature), the proposed solution utilizes a web camera along with a YOLOv8-based machine learning model to detect and classify infant sleep positions and identify early signs of distress. The system enables live video monitoring and provides real-time alerts to caregivers through Telegram Bot. This design improves infant safety, reduces parental workload, and introduces predictive intelligence that is missing in conventional cradle monitoring systems.

4.2 Block Diagram

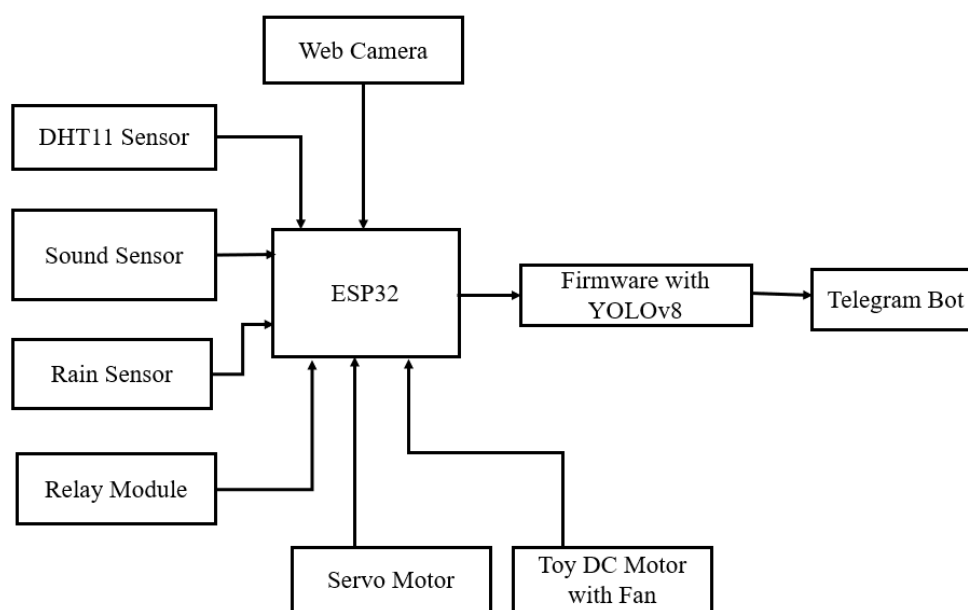


Figure 4.2 Block Diagram

The above Fig 4.2.1 illustrates the block diagram illustrates the overall working architecture of the AI-enhanced IoT Cradle Monitoring System, where the ESP32 microcontroller acts as the central processing unit. Various sensors such as the DHT11 sensor for temperature and humidity, the sound sensor for detecting crying, and the rain sensor for moisture or water detection are connected to the to continuously monitor the baby's environment. Web Camera is integrated to capture real-time images of the infant, which are processed using the YOLOv8-based firmware to identify unsafe sleeping

positions. The system includes a relay module that controls the toy DC motor with a fan for ventilation, while a servo motor is used to tilt or gently swing the cradle whenever needed., enabling caregivers to view live updates from anywhere. In case of abnormal conditions such as high temperature, crying, moisture detection, or unsafe posture the instant alert notifications are sent through a Telegram Bot, ensuring quick caregiver response. Overall, the diagram shows how all hardware and software components work together to provide a safe, intelligent, and automated cradle monitoring system.

4.3 Hardware Components

This section describes the major hardware components used in the proposed AI-Enhanced IoT-based cradle monitoring system. Each component plays a crucial role in sensing the infant's condition, capturing real-time environmental data, and enabling automated cradle responses. The integration of these components ensures effective monitoring, predictive alert generation, and smooth communication with cloud platforms.

4.3.1 ESP32 Microcontroller

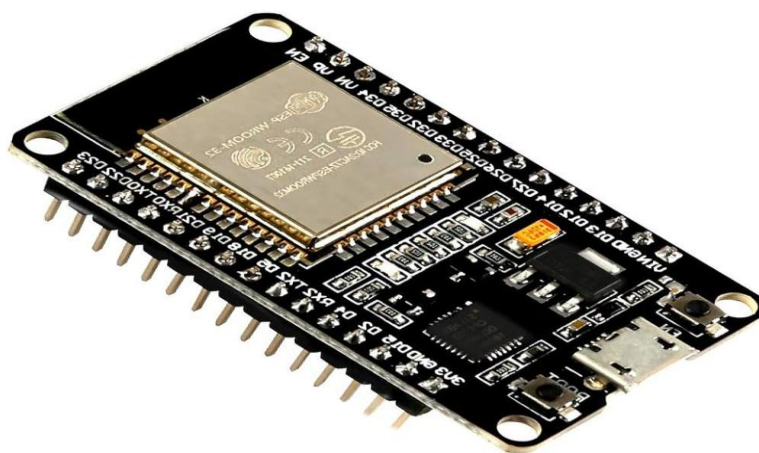


Figure 4.3.1 ESP32

In the above Fig 4.3.1 This section describes the major hardware components used in the proposed AI-Enhanced IoT-based cradle monitoring system. Each component plays a crucial role in sensing the infant's condition, capturing real-time environmental data, and enabling automated cradle responses. The integration of these components ensures effective monitoring, predictive alert generation, and smooth communication with cloud platforms.

4.3.2 Web Camera



Figure 4.3.2 Web Camera

The Web Camera is used for real-time video streaming and image capture. It provides visual monitoring of the infant, enabling AI-based predictions such as posture detection, movement tracking, and potential risk identification. In the above Fig 4.3.2, the Web Camera is connected to a computer system via USB, where the captured video is processed using machine learning algorithms. The processed results are then transmitted to the ESP32 module through serial communication (UART), which further handles alert generation and communication with cloud platforms for real-time monitoring.

4.3.3 DHT11 Sensor

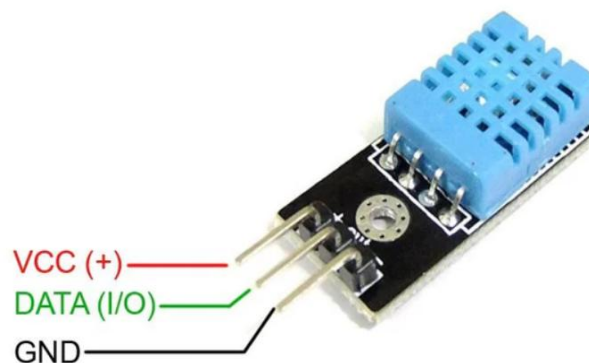


Figure 4.3.3 DHT11 Sensor

Fig 4.3.3 shows the DHT11 sensor monitors two essential comfort parameters: temperature and humidity around the cradle. The ESP32 receives these values and

evaluates whether the baby's environment is optimal. Abnormal conditions trigger instant alerts to caregivers.

4.3.4 Rain Sensor

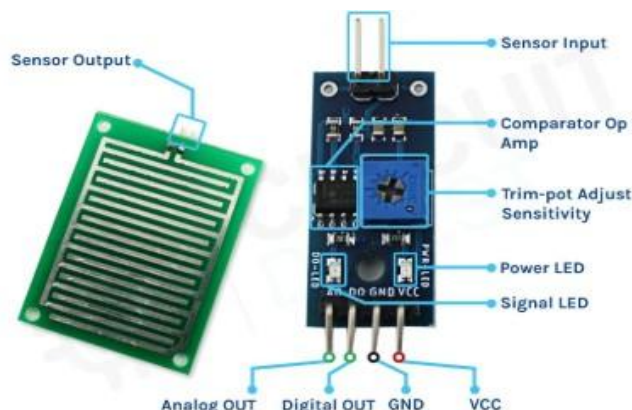


Figure 4.3.4 Sensor Sensor

The rain sensor is repurposed to detect wetness on the cradle mattress, indicating possible diaper leakage or spills. In the above Fig 4.3.4 When moisture is detected, the sensor sends an analog signal to the ESP32, which then notifies parents through Telegram alerts.

4.3.5 Sound Sensor



Figure 4.3.5 Sound Sensor

Fig 4.3.5 shows the sound sensor detects baby cries or distress sounds. It helps identify discomfort or emergencies by continuously monitoring sound intensity. When crying exceeds a threshold, the ESP32 activates notifications and can trigger servo-based rocking responses.

4.3.6 Servo Motor



Figure 4.3.6 Servo Motor

A servo motor is installed to generate controlled rocking motion for the cradle. In the above Fig 4.3.6 When sensors detect discomfort (crying or movement), the ESP32 activates the servo motor to gently rock the cradle as a soothing action.

4.3.7 Toy DC Motor with Fan

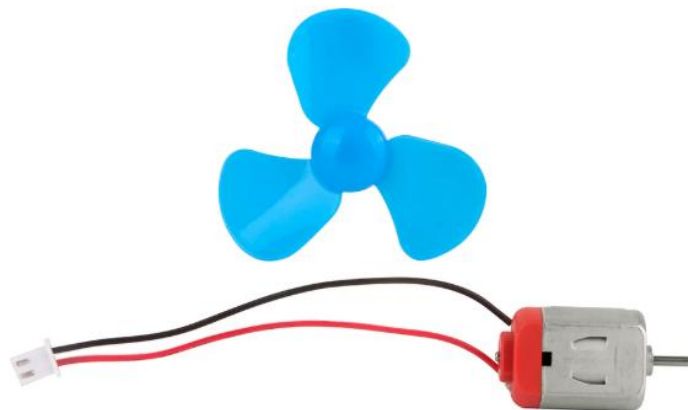


Figure 4.3.7 Toy DC Motor with Fan

Fig 4.3.7 explains a small toy DC motor attached to a fan helps circulate air and maintain a comfortable environment for the infant. The motor is controlled via a relay module and is activated when temperature crosses a threshold value.

4.3.8 Relay Module

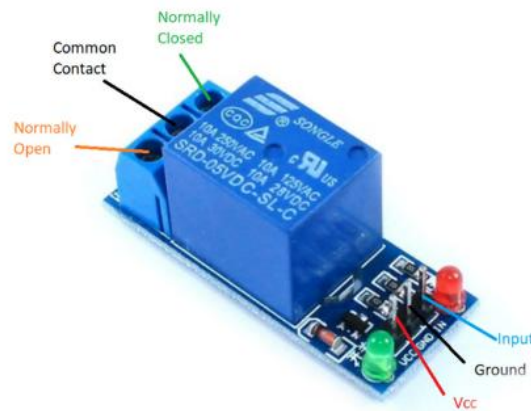


Figure 4.3.8 Relay Module

The relay module is used to operate high-current components such as the toy DC motor. In the above Fig 4.3.8 It ensures electrical isolation and safe switching controlled by the ESP32. When temperature increases, the ESP32 activates the relay to turn ON the fan.

4.3.9 Jumper Wires



Figure 4.3.9 Jumper Wires

In Fig 4.3.9 Jumper wires are flexible, insulated wires with solid metal tips (male or female) on either end. They are used for temporary or semi-permanent connections on breadboards or between header pins.

4.3.10 Breadboard

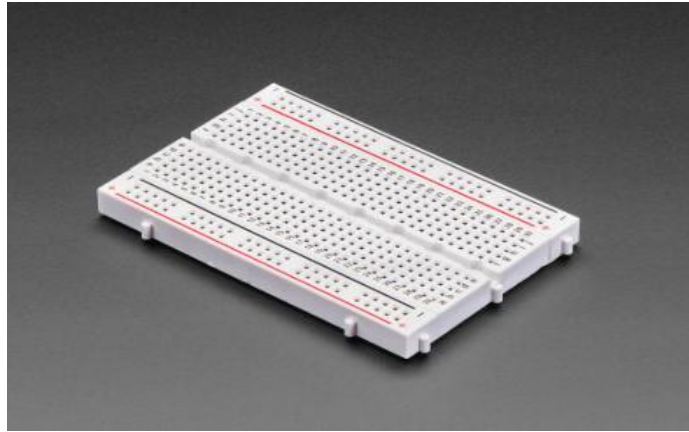


Figure 4.3.10 Breadboard

A breadboard is a plastic board with interconnected holes used to build and test electronic circuits without soldering. In above the Fig 4.3.10 the system, it plays a vital role in connecting sensors, ESP32, the relay module, and other components temporarily during prototyping.

- Assemble circuits quickly
- Test and modify your design without permanent connections
- Avoid soldering, which saves time and components
- Safely experiment with connections before final deployment.

4.4 Software Components

4.4.1 YOLOv8 Model Integration

YOLOv8 (You Only Look Once version 8) is used as the core machine learning model for real-time object detection. In this project, it is trained to identify infant sleep positions such as safe and unsafe postures. The model processes video frames from the Web Camera and detects patterns like face orientation, body movement, and sleeping posture. Its high speed and accuracy make it suitable for continuous monitoring and quick response systems.

4.4.2 Roboflow Platform

Roboflow is used for dataset preparation, annotation, and model training. Images of different infant sleep positions are uploaded, labeled, and preprocessed using this platform. It also helps in augmenting data (such as rotation, flipping, resizing) to improve

model performance. After training, the model is exported and integrated into the detection system.

4.4.3 Visual Studio Code (VS Code)

Visual Studio Code is the development environment used to write, test, and debug the project code. It supports Python programming, integrates with machine learning libraries, and allows smooth execution of the detection system. Extensions in VS Code help manage dependencies, run scripts, and monitor outputs efficiently.

4.4.4 Machine Learning Algorithm for Detection

The system uses a **supervised machine learning based object detection algorithm**, specifically the YOLOv8 (You Only Look Once) model. YOLOv8 is a deep learning-based computer vision algorithm that performs real-time detection by analyzing video frames and identifying objects in a single pass. In this project, the model is trained using labeled images of infant sleep positions to classify them as **safe or unsafe**. The algorithm uses **Convolutional Neural Networks (CNNs)** to extract features such as body orientation, face position, and movement patterns. Based on this analysis, the system predicts potential risks and triggers alerts when unsafe conditions are detected.

4.4.5 Telegram Bot for Alerts and Notifications

A Telegram Bot is integrated to provide instant alerts to caregivers. When an unsafe sleeping position or abnormal condition is detected, the system sends a notification message along with images or status updates. This ensures real-time monitoring even when the caregiver is not physically present.

4.5 Circuit Diagram

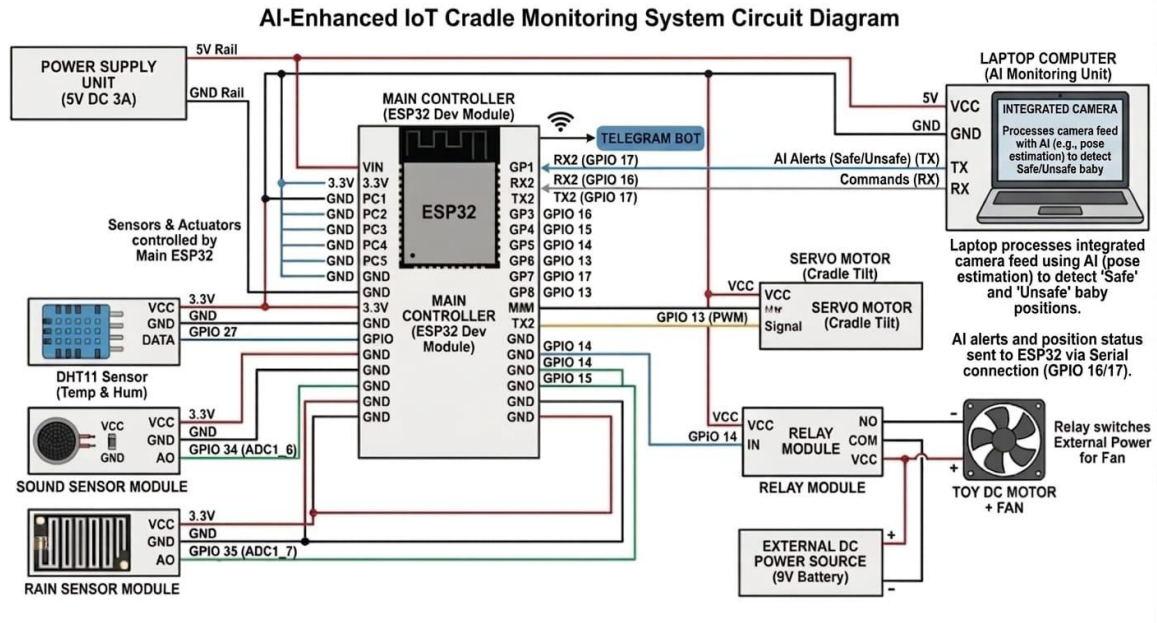


Figure 4.5 Circuit Diagram

The circuit diagram illustrates the complete working structure of the AI-Enhanced IoT Cradle Monitoring System, where the ESP32 acts as the main controller responsible for gathering data, running firmware, and handling communication processes. In the above Fig 4.5, different sensors such as the DHT11 sensor, sound sensor, and rain sensor are connected to the ESP32's GPIO pins to continuously monitor the baby's environment, including temperature, humidity, noise levels, and moisture. The servo motor, connected through a PWM pin, is used to tilt the cradle automatically based on detected conditions or alerts. The relay module controls a Toy DC Motor with a fan, which can turn ON or OFF depending on safety alerts or environmental readings.

A Web Camera is used for visual monitoring of the infant. The captured video is processed on a computer system using the Roboflow platform and Visual Studio Code, where a trained YOLOv8 model is used to detect baby sleep positions as safe or unsafe. Based on the detection results, alerts are generated through the Telegram bot application and communicated to the ESP32 system for further action.

4.6 Hardware Connection

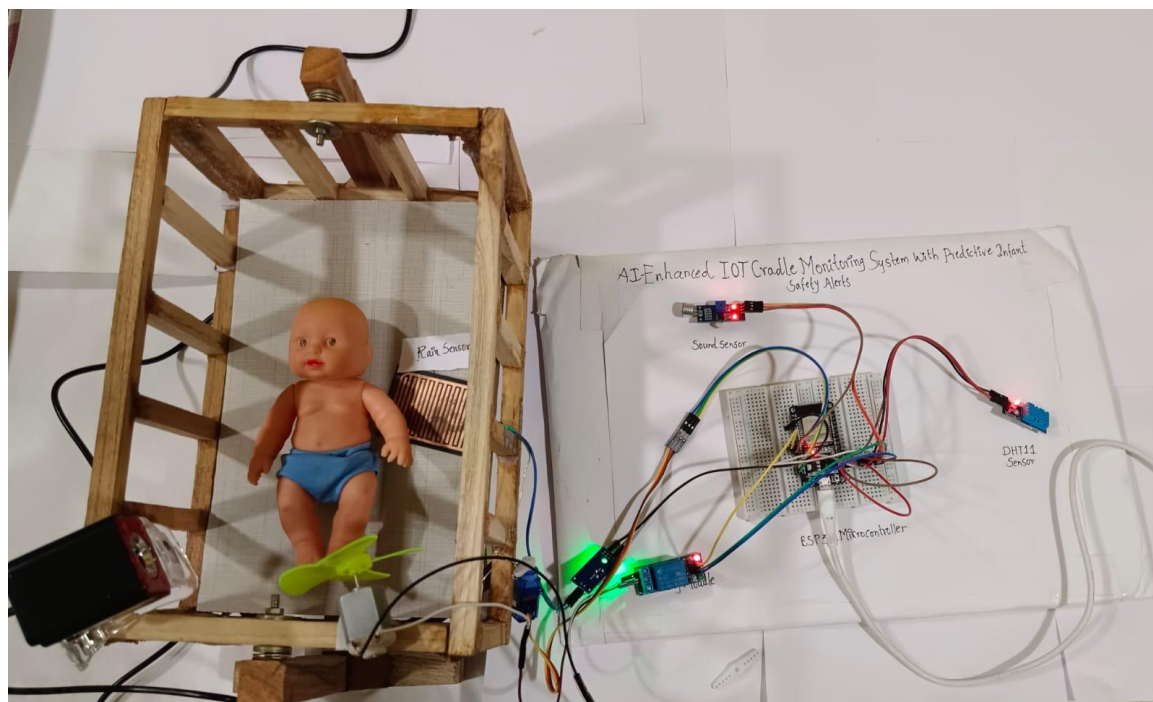


Figure 4.6 Hardware Connection

In above Figure 4.6 The hardware framework of the AI-Enhanced IoT Cradle Monitoring System is designed around a central microcontroller that interfaces with multiple sensory inputs and mechanical actuators to ensure automated infant safety and comfort.

Central Processing Unit: The core of the system is an ESP32 Microcontroller, mounted on a breadboard. It acts as the primary IoT gateway, processing sensor data, communicating with the Telegram bot via Wi-Fi, and triggering the appropriate hardware responses.

4.6.1 Environmental and State Sensors (Inputs):

- **DHT11 Sensor:** Interfaced directly with the ESP32, this sensor continuously monitors the ambient temperature and humidity levels within the cradle's vicinity to ensure optimal environmental conditions.
- **Sound Sensor:** Connected to the microcontroller to capture acoustic signals. It is calibrated to detect specific decibel thresholds corresponding to a baby's cry, triggering the automated soothing mechanisms.

- **Rain Sensor:** Repurposed as a diaper wetness detector, the sensor pad is placed flat on the cradle bed. It detects liquid conductivity, sending an interrupt to the ESP32 the moment dampness occurs.

4.6.2 Actuators and Control Mechanisms (Outputs):

- **Relay Module:** Used as an electrically operated switch, allowing the low-power ESP32 signals to safely control higher-power components in the system.
- **Cooling Fan (DC Motor):** A DC motor fitted with a fan blade is mounted inside the cradle area. This is programmed to activate automatically if the DHT11 sensor registers temperatures exceeding the defined comfort threshold.
- **Cradle Tilting Mechanism (Servo Motor):** A servo motor is mechanically linked to the wooden cradle's pivot system. When the sound sensor detects crying, the ESP32 commands the servo to initiate a gentle rocking or tilting motion to soothe the infant.
- **Vision Module:** A camera module is mounted securely to the side frame of the cradle, angled downward to maintain a clear, unobstructed view of the infant's sleep area. This camera provides the continuous live video feed required for the AI-based Roboflow object detection model to analyze the baby's posture.

All components are interconnected using standard jumper wires, creating a cohesive, automated, and highly responsive smart cradle prototype.

4.7 System Operational Flowchart

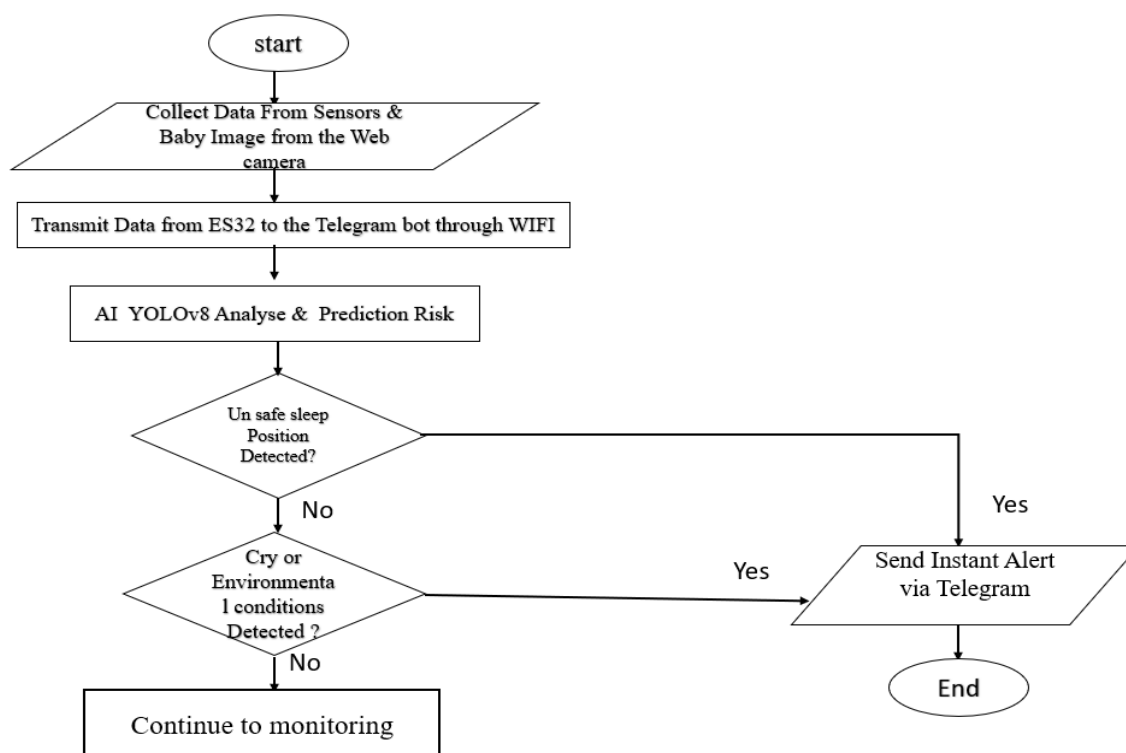


Figure 4.7 System Operational Flowchart

The above Fig 4.7 flowchart illustrates the step-by-step operational logic of the AI-Enhanced IoT Cradle Monitoring System. The process begins with the data collection phase, where environmental data from sensors such as the DHT11, rain sensor, and sound sensor connected to the ESP32 are continuously monitored. At the same time, visual data is captured using a Web Camera. The video data is processed on a computer system using a trained YOLOv8 model to detect the infant's sleep position as safe or unsafe.

The system follows a two-level decision-making process. First, it checks whether an unsafe sleep position is detected. If yes, an immediate alert is sent to the caregiver through the Telegram Bot. If no unsafe position is detected, the system proceeds to check environmental conditions such as baby cry or unfavourable temperature and humidity levels. If any abnormal condition is identified, an alert is generated. If all conditions are normal, the system continues monitoring in a loop.

This approach ensures continuous real-time monitoring, prioritizing critical conditions like unsafe sleep positions and providing timely alerts to ensure infant safety.

CHAPTER-5

RESULT & DISCUSSION

5.1 Detection of Safe and Unsafe Sleep Positions Using Roboflow



Figure 5.1.1 Safe Sleep Position using Roboflow

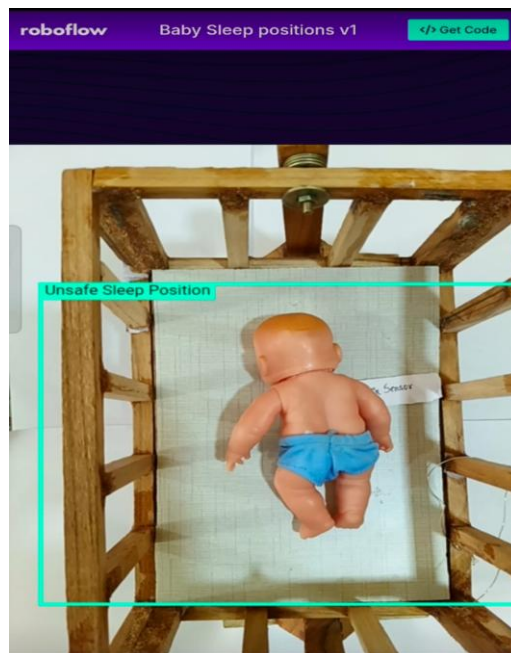


Figure 5.1.2 Unsafe Sleep Position Roboflow

Detection of Safe and Unsafe Sleep Positions in Roboflow is performed by training a custom dataset of infant images labelled as safe and unsafe sleeping positions as shown in the above figures as 5.1.1 and 5.1.2. The images are annotated and preprocessed using Roboflow tools, including resizing and augmentation techniques to improve model accuracy. The trained dataset is then used to develop a YOLOv8 model, which can accurately identify the baby's posture in real-time and classify it as safe or unsafe during monitoring.

5.2 Real-Time Baby Sleep Position Detection Using Web Camera



Figure 5.2.1 Real-Time Safe Sleep Position

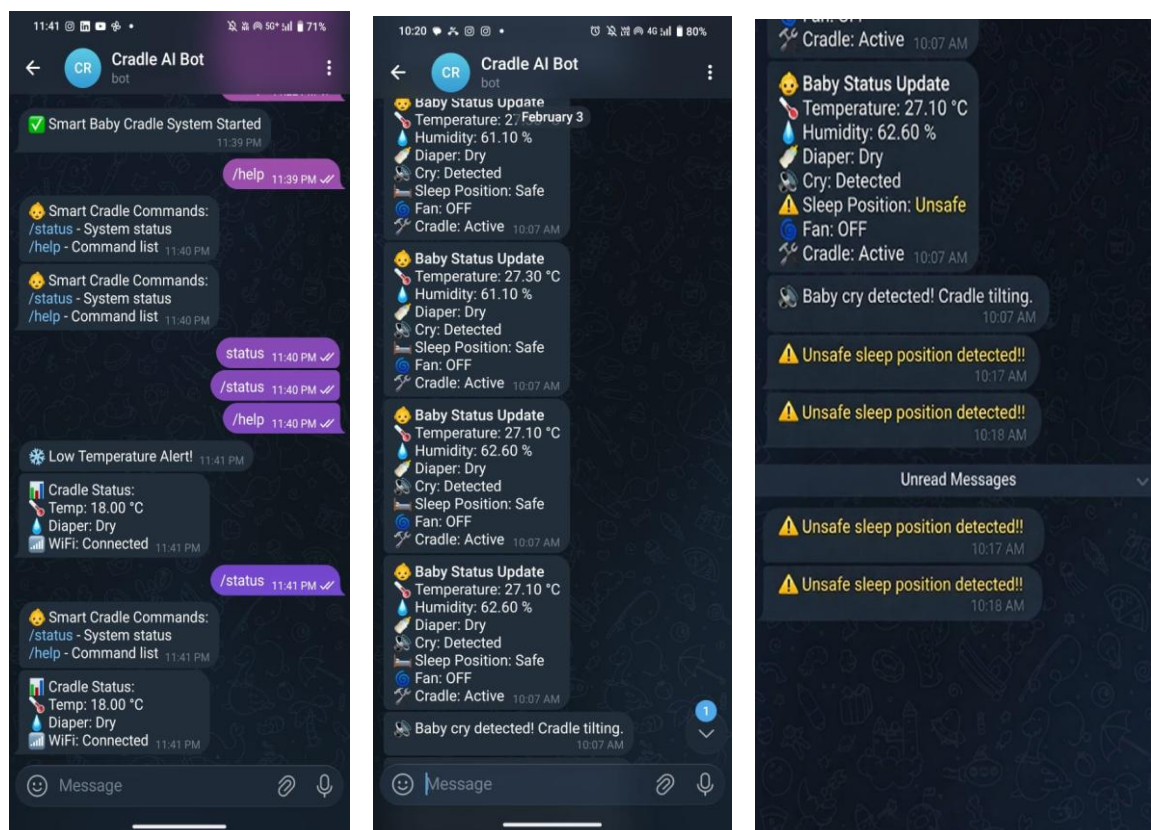
The real-time safe sleep position is as shown in the above fig 5.2.1 of the baby is identified when the infant is lying on its back (supine position) with the face clearly visible and unobstructed. In this position, the head is oriented upward, ensuring that the airway remains open and reduces the risk of suffocation or breathing difficulties. The system detects this posture using a webcam and classifies it as “SAFE” with a confidence score, providing reassurance to caregivers. This position aligns with recommended pediatric safety guidelines, as it minimizes the chances of Sudden Infant Death Syndrome (SIDS) and ensures proper airflow for the baby during sleep.



Figure 5.2.2 Real-Time Unsafe Sleep Position

The Above Figure 5.2.2 shows an unsafe sleep position occurs when the baby is lying on its stomach (prone position) or in a posture where the face is turned downward or obstructed. In such cases, the airway may be partially or completely blocked, increasing the risk of suffocation and breathing problems. The system analyses live video input from the webcam to detect these risky orientations and labels them as “UNSAFE” with a confidence score. Once detected, alerts can be generated to notify caregivers immediately, allowing quick action to reposition the baby into a safer sleeping posture.

5.3 Real-Time Alerts using Telegram Bot



Figures 5.3.1–5.3.3: Real-Time Alert Notifications

Based on the above Figures 5.3.1, 5.3.2 & 5.3.3 the Cradle AI Bot is a smart, automated baby monitoring system integrated with Telegram. It provides both proactive real-time alerts and on-demand status updates to keep caregivers informed about the baby's environment and well-being.

Here is a brief overview of its real-time alerting capabilities: Proactive Real-Time Alerts

The bot pushes immediate notifications to the user when specific events are triggered by the hardware sensors:

Cry Detection & Response: It detects when the baby is crying and notifies the user with messages like "Baby cry detected! Cradle tilting," indicating that the system not only detects the cry but also automatically activates the cradle to soothe the baby.

Hygiene Alerts: It monitors for moisture and sends a "Diaper wetness detected!" alert so caregivers know exactly when a change is needed.

Safety Warnings: It monitors the baby's posture and issues a highly visible warning stating "Unsafe sleep position detected!!" (accompanied by warning icons) to ensure the baby's physical safety.

On-Demand Status Updates:

Users can interact with the bot using commands like /status to receive a comprehensive "Baby Status Update" dashboard at any time.

This update includes: Environmental Metrics: Current Temperature (e.g., 32.20 °C) and Humidity (e.g., 51.50%).

Baby's Condition: Current diaper status (Wet/Dry), cry detection status, and sleep position safety (Safe/Unsafe).

System Status: Indicates whether the connected Fan is ON/OFF and if the Cradle is currently Active.

Overall, the Telegram bot acts as a centralized, remote control hub that ensures parents receive instant, critical updates about their baby's comfort and safety directly to their phones.

CHAPTER-6

CONCLUSION & FUTURE SCOPE

6.1 Final Conclusion

The "AI-Enhanced IoT Based Cradle Monitoring System with Predictive Infant Safety Alerts" successfully fulfills its primary objective of creating a comprehensive, automated, and predictive safety environment for infants. By seamlessly integrating an advanced computer vision model with a robust IoT hardware ecosystem, the project transcends traditional baby monitoring systems.

The implementation of the AI vision module proved highly effective in continuously analyzing the infant's sleep posture, reliably distinguishing between safe (supine) and high-risk (prone) positions in real-time. This predictive capability acts as a crucial preventative measure against sleep-related hazards, such as SIDS, by identifying danger the moment a physical transition occurs.

Beyond visual monitoring, the system's sensory network—driven by the ESP32 microcontroller—demonstrated excellent responsiveness to the infant's immediate needs and environmental conditions. The successful deployment of sound and moisture sensors, coupled with the DHT11 module, ensured that discomfort (crying, wet diapers, or high temperatures) was instantly detected. Furthermore, the system proved its ability to transition from passive monitoring to active intervention by automatically triggering the mechanical cradle-rocking servo and cooling fan to soothe the infant.

Ultimately, the true value of this system is realized through its user-centric communication framework. The low-latency Telegram bot integration ensures that caregivers are never disconnected from the infant's status. By providing instant emergency alerts for critical events and comprehensive on-demand environmental updates, the system grants parents unparalleled peace of mind. This project successfully demonstrates that the convergence of edge AI, IoT sensor networks, and automated mechanical design can effectively transform a traditional cradle into a proactive, intelligent guardian of infant safety.

6.2 Future Scope

While the current prototype successfully demonstrates the core functionalities of an AI-enhanced infant monitoring system, there are several promising avenues for future development that can further elevate its capabilities, reliability, and commercial viability:

Advanced Night Vision Capabilities: The current computer vision model relies on ambient lighting. Integrating an Infrared (IR) camera module would allow the AI to seamlessly monitor the infant's sleep posture in complete darkness without disturbing the baby's sleep cycle.

Non-Contact Vital Sign Monitoring: Future iterations could incorporate advanced sensor technologies, such as low-power radar (e.g., mmWave sensors), to detect micro-movements. This would allow the system to monitor the infant's respiratory rate and heart rate continuously without attaching any wires to the baby, offering an even stronger safeguard against Sudden Infant Death Syndrome (SIDS).

Dedicated Mobile Application and Cloud Analytics: While the Telegram bot is highly responsive for a prototype, developing a dedicated, cross-platform mobile application (iOS/Android) would provide a richer user interface. Additionally, pushing the sensor data to a cloud database (like AWS or Google Cloud) would allow the system to track and analyze the baby's sleep patterns over time, providing parents with actionable insights into their child's sleep health.

Enhanced Actuation and Soothing Mechanisms: The mechanical design can be refined by replacing the standard servo motor with a high-torque stepper motor, ensuring a much smoother and whisper-quiet rocking motion. Furthermore, integrating a speaker module to automatically play white noise or pre-recorded lullabies when the sound sensor detects crying would create a more holistic self-soothing environment.

Facial Expression and Emotion Analysis: By upgrading the edge computing hardware (e.g., to a Raspberry Pi or an AI accelerator like a Google Coral TPUs), the vision model could be expanded to include facial emotion recognition. This would allow the system to detect subtle signs of pain, illness, or distress even before the infant begins to cry.

Power Redundancy: Integrating a customized Battery Management System (BMS) with a rechargeable power bank would ensure the cradle remains fully operational and continues to send critical alerts during sudden power outages.

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Mapping of Project Outcomes (PrOs) with Program Outcomes

(POs) and Program Specific Outcomes (PSOs)

Title of the Project: **AI-Enhanced IoT Cradle Monitoring System with Predictive Infant Alerts**

Type of the Project: (Multiple options can be chosen)

Product-based	☒
Modern tool usage	☒
Application-based	☒
Research-based	☐

Upon completion of this Project, we are able to

Project Outcome 1	Successfully implemented real-time monitoring using cameras and sensors.
Project Outcome 2	Alerts are automatically sent via Telegram/Blynk for immediate action.
Project Outcome 3	Baby's safe and unsafe positions are detected for better monitoring. Reduced manual supervision through an efficient and automated system.

PrOs Mapped with POs and PSOs

PrO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
PrO 1	3	3	3	3	3	2	1	-	2	2	1	2	3	3
PrO 2	3	3	3	3	3	2	1	-	3	3	2	2	3	3
PrO 3	2	2	3	3	3	2	1	-	3	3	2	3	3	3
Avg	2.66	2.66	3	3	3	2	1	-	2.66	2.66	1.66	2.33	3	3

1: LOW 2: MODERATE 3: HIGH



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Mapping of Project with Sustainable Development Goals (SDG)

Title of the Project: **AI-Enhanced IoT Cradle Monitoring System with Predictive Infant Alerts**

Project mapping with SDGs: **SDG4, SDG9, SDG12**

Justification on Project Mapping with SDG:

SDG No.	SDG Goal	JUSTIFICATION
SDG4	Quality Education	Promotes learning of advanced technologies like YOLO, sensors, and real-time data processing, enhancing technical education and skill development.
SDG9	Industry, Innovation and Infrastructure	Demonstrates innovation by integrating AI and IoT into practical infrastructure for home safety and agriculture, fostering smart solutions.
SDG12	Sustainable Cities and Communities	Reduces crop loss and resource wastage, encouraging sustainable agricultural practices and responsible use of resources.